fpga_cpu: Spartan-3 Starter Board expansion (rough). Serial and the keyboard are on the board itself (DB9: RXD T13, TXD R13; PS/2: M15/M16).

A2 expansion connector (UG130 table 13-3)
4 E6 I2S_BCK 5 D5 I2S_LRCK 6 C5 I2S_DATA
7 D6 MS_CLK 8 C6 MS_DATA
9 E7 SD_CS 10 C7 SD_SCK 11 D7 SD_MOSI
12 C8 SD_MISO 13 D8 SD_CD
14 C9 PAD_LATCH 15 D10 PAD_CLK
16 A3 PAD1_DATA 17 B4 PAD2_DATA
18 A4 WIFI_CS 19 B5 WIFI_SCK 20 A5 WIFI_MOSI 21 B6 WIFI_MISO
22 B7 WIFI_HS 23 A7 WIFI_EN 24 B8 WIFI_BOOT 25 A8 WIFI_TXD 26 A9 WIFI_RXD
27-35 spare: 36-40 configuration: leave open
2 +5V (VU), 3 +3.3V (VCCO), 1 GND

Sound: I2S (64 BCK a frame, 48.828 Hz) into a PCM5102A; SCK tied low: the DAC's PLL makes its clock from BCK. FMT low = I2S. XSMT high = unmuted. Line out through a 470R / 2.2nF filter.

PS/2 mouse (the keyboard uses the board's own PS/2 port, M15/M16). Open drain both ways: the FPGA drives low or lets go. BSS138 level shifters (gate at 3.3 V): 10k pull-ups on both sides. Not 5 V tolerant!

WiFi: ESP32-WROOM-32E (e.g. ESP-AT firmware) on its own pins, not the SD bus: shared, the ESP's MISO would be GPIO12, a strap the SD pull-up would hold high (1.8 V flash: no boot). VSPI slave: SCK IO18, MOSI IO23, MISO IO19, CS IO5; IO4 = handshake/interrupt to the FPGA; EN (reset) and IO0 (download mode) driven by the FPGA (open drain; pull-ups, EN RC). UART1 (IO16 in, IO17 out) to the FPGA too; UART0 on J8 for flashing. Own AMS1117 from +5V: the module draws ~500 mA when transmitting. One SPI engine in the FPGA serves SD and WiFi.

microSD in SPI mode: DAT3 = CS, CMD = MOSI, DAT0 = MISO; DAT1/DAT2 pulled up (unused); card detect to the FPGA, 47k pull-ups, 33R on the clock.

NOR flash ROM, executed in place (a resident toolbox: no RAM spent on it). 512 KB x 8, 3.3 V (SST39VF040; the symbol is the 5 V SST39SF040, same JEDEC pinout). For 1 MB (SST39VF080, TSOP-40) ROM_A19 is its A19. CE# pulled up: off while the FPGA configures. Writes (programming) use the JEDEC command sequences through the same bus, with a long WE# pulse.

NES pads (pins: 1 GND, 2 CLK, 3 LATCH, 4 DATA, 7 VCC), powered at 3.3 V (the pad's 4021 runs from 3 V, so DATA needs no level shifting). An SNES pad uses the same three signals (16 bits). Pull-ups: no pad reads 'none'.

A1 expansion connector (UG130 table 13-2): shared with the board's SRAM
5-19 odd: IC10 IO0-IO7 (sram_d[7:0]): A0-A17: pins 6, 8, 10, 12, 14, 24, 23, 26, 25, 28, 27, 30, 29, 32, 31, 34, 33, 35; 16 WE#, 18 OE#
Free: 4 N8 ROM_CE, 21 M7 ROM_A18, 22 M10 ROM_A19, 20 P9 (DOUT: unused)
36-40: JTAG (TMS, TCK, TDO): do not connect

A1 pin 22 (M10): A19 for a 1 MB part (SST39VF080): unused with 512 KB.
J3 needs a mini-DIN-6 footprint (e.g. CUI MD-60S): none in KiCad's library.

rough: sound, PS/2 mouse, microSD, NES pads, ESP32 WiFi (A2); NOR flash ROM on the SRAM bus (A1)

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Title: fpga_cpu Spartan-3 expansion

Size: A3	Date:	Rev: 0.1
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